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gayathri@Gayathri: ~/OSLab
cdgayathri@Gayathri:~$ cd OSLab
gayathri@Gayathri:~/OSLab$ nano memory.c
gayathri@Gayathri:~/OSLab$ gcc memory.c -o memory
gayathri@Gayathri:~/OSLab$ ls
OS_Lab  a.out  memory.c  pipes  program10.c  progr
am2.c  program4.c  program6.c  server.c.save  process  program11  progr
Practical6  hello  one.c  signals
am3  program5  program7
Signals  hello.c  permission  process.c  program11.c  progr
am3.c  program5.c  program7.c
Signals.c  memory  permission.c  program10  program2  progr
am4  program6  sample.txt
gayathri@Gayathri:~/OSLab$ ./memory
PID          : 496
Code address  : 0x5611e9fc71c9
Global address : 0x5611e9fca010
Static address : 0x5611e9fca014
Heap address   : 0x561224d58010
Stack address  : 0x7ffccbdb104c

Process is running...
Press Ctrl+C to terminate.
^C
gayathri@Gayathri:~/OSLab$ |
```

```
gayathri@Gayathri:~$ cat /proc/496/maps
5611e9fc6000-5611e9fc7000 r--p 00000000 08:30 14505 /home/gayathri/OSLab/memory
5611e9fc7000-5611e9fc8000 r-xp 00001000 08:30 14505 /home/gayathri/OSLab/memory
5611e9fc8000-5611e9fc9000 r--p 00002000 08:30 14505 /home/gayathri/OSLab/memory
5611e9fc9000-5611e9fca000 r--p 00002000 08:30 14505 /home/gayathri/OSLab/memory
5611e9fca000-5611e9fcb000 rw-p 00003000 08:30 14505 /home/gayathri/OSLab/memory
561224d58000-561224d79000 rw-p 00000000 00:00 0 [heap]
7e46bb200000-7e46bb228000 r--p 00000000 08:30 13647 /usr/lib/x86_64-linux-gnu/libc.so.6
7e46bb228000-7e46bb3c0000 r-xp 00028000 08:30 13647 /usr/lib/x86_64-linux-gnu/libc.so.6
7e46bb3c0000-7e46bb40e000 r--p 001c0000 08:30 13647 /usr/lib/x86_64-linux-gnu/libc.so.6
7e46bb40e000-7e46bb412000 r--p 0020d000 08:30 13647 /usr/lib/x86_64-linux-gnu/libc.so.6
7e46bb412000-7e46bb414000 rw-p 00211000 08:30 13647 /usr/lib/x86_64-linux-gnu/libc.so.6
7e46bb414000-7e46bb421000 rw-p 00000000 00:00 0
7e46bb48d000-7e46bb490000 rw-p 00000000 00:00 0
7e46bb497000-7e46bb499000 rw-p 00000000 00:00 0
7e46bb499000-7e46bb49d000 r--p 00000000 00:00 0
7e46bb49d000-7e46bb4a1000 r--p 00000000 00:00 0
7e46bb4a1000-7e46bb4a2000 r--p 00000000 08:30 13644 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
7e46bb4a2000-7e46bb4d1000 r-xp 00001000 08:30 13644 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
7e46bb4d1000-7e46bb4dc000 r--p 00030000 08:30 13644 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
7e46bb4dc000-7e46bb4de000 r--p 0003b000 08:30 13644 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
7e46bb4de000-7e46bb4df000 rw-p 0003d000 08:30 13644 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
7e46bb4df000-7e46bb4e0000 rw-p 00000000 00:00 0
7ffccbd92000-7ffccbdb3000 rw-p 00000000 00:00 0 [stack]
gayathri@Gayathri:~$ pmap 496
496: ./memory
00005611e9fc6000 4K r---- memory
00005611e9fc7000 4K r-x-- memory
00005611e9fc8000 4K r---- memory
00005611e9fc9000 4K r---- memory
00005611e9fca000 4K rw--- memory
0000561224d58000 132K rw--- [ anon ]
00007e46bb200000 160K r---- libc.so.6
00007e46bb228000 1632K r-x-- libc.so.6
```

OUTPUT:-

```
gayathri@Gayathri: ~$ cat /proc/cpuinfo
```

```
VmLib:      1828 kB  
VmPTE:      48 kB  
VmSwap:     0 kB  
HugetlbPages:    0 kB  
CoreDumping:   0  
THP_enabled:   1  
untag_mask: 0xfffffffffffffff  
Threads:       1  
SigQ:        1/63103  
SigPnd: 0000000000000000  
ShdPnd: 0000000000000000  
SigBlk: 0000000000000000  
SigIgn: 0000000000000000  
SigCgt: 0000000000000000  
CapInh: 0000000000000000  
CapPrm: 0000000000000000  
CapEff: 0000000000000000  
CapBnd: 000001ffffffffffff  
CapAmb: 0000000000000000  
NoNewPrivs:    0  
Seccomp:        0  
Seccomp_filters: 0  
Speculation_Store_Bypass: thread vulnerable  
SpeculationIndirectBranch: conditional enabled  
Cpus_allowed:   fff  
Cpus_allowed_list: 0-11  
Mems_allowed:   00000000,00000000,00000000,00000000,00000000,00000000,00000000,00000000,00000000,00000000,00000000,00000000,00000000,00000000,  
00000000,00000000,00000000,00000000,00000000,00000000,00000000,00000000,00000000,00000000,00000000,00000000,00000000,00000000,00000000,00000000,  
0,00000000,00000000,00000000,00000001  
Mems_allowed_list: 0  
voluntary_ctxt_switches: 193  
nonvoluntary_ctxt_switches: 0
```

```
gayathri@Gayathri:~$ |
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